Introduction to Logic: Lecture 1

Kyle Richardson

October 20 2026

Course Overview

What this course is about...

Logic: the study of reasoning, inference, and how one constructs correct arguments.

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"You can access the Internet from campus only if **you are a linguistics major** or you are **not a freshman**. Mary **can access the Internet**. So, Mary is **a linguistics major** or **not a freshman**."

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The same argument in symbols

$$P_a \rightarrow (P_l \vee \neg P_f), \quad P_a \quad \therefore \quad P_l \vee \neg P_f$$

Logical reasoning shows up everywhere...

Solving puzzles

The butler or the gardener did it. The butler didn't. *So, the gardener did.*

$$P_b \vee P_g, \quad \neg P_b \quad \therefore \quad P_g$$

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Medicine

If it's strep, the test is positive. The test is negative. *So, it's not strep.*

$$P_s \rightarrow P_t, \quad \neg P_t \quad \therefore \quad \neg P_s$$

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Law

You may end the lease early only if the flat is unsafe. The flat is safe. *So, you may not.*

$$P_e \rightarrow P_u, \quad \neg P_u \quad \therefore \quad \neg P_e$$

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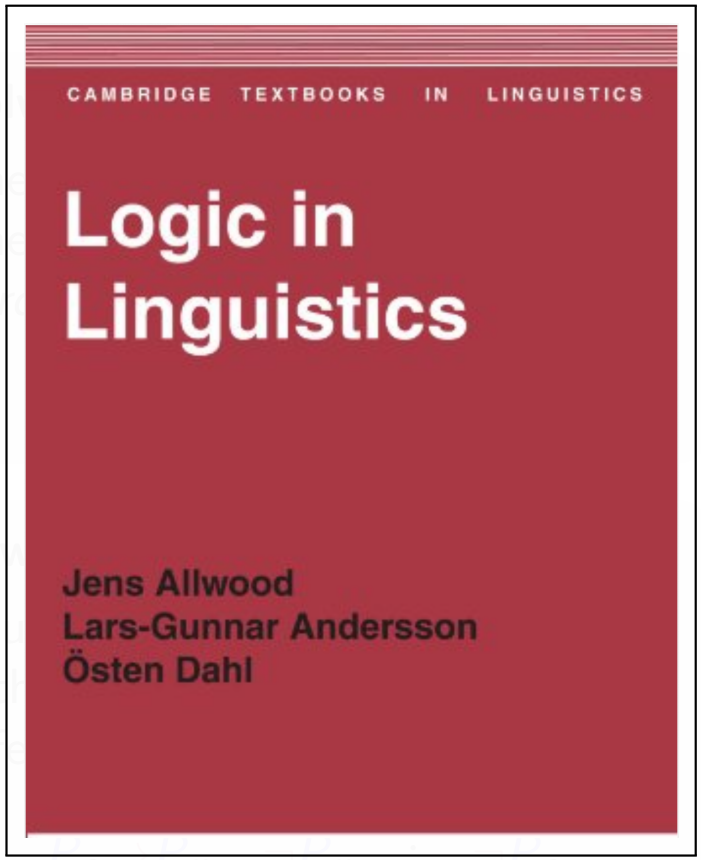
$$P_e \rightarrow P_u, \quad \neg P_u \quad \therefore \quad \neg P_e$$

Databases

Books on Woolf or Joyce, but not on poetry. `(Woolf OR Joyce) AND NOT poetry`

$$(P_w \vee P_j) \wedge \neg P_p$$

Logical reasoning shows up everywhere...



THE BULLETIN OF SYMBOLIC LOGIC
Volume 7, Number 2, June 2001

ON THE UNUSUAL EFFECTIVENESS OF LOGIC IN COMPUTER SCIENCE

JOSEPH Y. HALPERN, ROBERT HARPER, NEIL IMMERMAN, PHOKION G. KOLAITIS, MOSHE Y. VARDI, AND VICTOR VIANU

§1. **Introduction and overview.** In 1960, E. P. Wigner, a joint winner of the 1963 Nobel Prize for Physics, published a paper titled *On the Unreasonable Effectiveness of Mathematics in the Natural Sciences* [61]. This paper can be construed as an examination and affirmation of Galileo's tenet that "The book of nature is written in the language of mathematics". To this effect, Wigner presented a large number of examples that demonstrate the effectiveness of mathematics in accurately describing physical phenomena. Wigner viewed these examples as illustrations of what he called *the empirical law of epistemology*, which asserts that the mathematical formulation of the laws of nature is both appropriate and accurate, and that mathematics is actually the *correct* language for formulating the laws of nature. At the same time, Wigner pointed out that the reasons for the success of mathematics in the natural sciences are not completely understood; in fact, he went as far as asserting that "... the enormous usefulness of mathematics in the natural sciences is something bordering on the mysterious and there is no rational explanation for it."

In 1980, R. W. Hamming, winner of the 1968 ACM Turing Award for Computer Science, published a follow-up article, titled *The Unreasonable*

What this course is about...

A formal language built from a few symbols:

$$P \qquad \neg \qquad \wedge \qquad \vee \qquad \rightarrow$$

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$$P \qquad \neg \qquad \wedge \qquad \vee \qquad \rightarrow$$

Syntax & semantics

What do formulas look like, and what do they mean?

Inference

How do we get from premises to a conclusion?

Tools we will learn along the way

- Understanding and building logics requires many mathematical tools.

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$\{a, b, c\}$

Sets

linguistics majors

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$R(a, b)$

Relations

loves, *older than*

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$\{a, b, c\}$	$R(a, b)$	$f : A \to B$
Sets	**Relations**	**Functions**
linguistics majors	*loves*, *older than*	*capital of*

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There is not just one logic, but many:

Classical	**Modal** $\Box \Diamond$	**Probabilistic**
and, or, not, if	*must, might*	*likely, probably*

What this course is not about...

This is not a mathematics class.

- We focus on *understanding* the techniques, not on proving purely mathematical theorems about them.
- We focus on applications to *linguistics and language*.

Thematic outline of course

Basic techniques

1. Course overview, set theory
2. Relations, functions

Classical logic

Propositional logic

3. Syntax
4. Semantics

Predicate logic

5. Syntax
6. Semantics
7. Proof methods

Extensions

8. Proofs/Modal logic
9. Discrete probability
10. Information theory

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- **Course evaluation**: Two exams: mid-term (50%) and final exam (50%), weekly readings and homework (not graded).

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- **Expectation**: Read weekly readings, go carefully through lecture slides and homework, attend tutor sessions (*as needed*).

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Questions?

(naïve) set theory

What is set theory about?

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Formal analysis requires talking about *groups of things*:

L	F	A
linguistics majors	freshmen	Internet users

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Set theory: the mathematical study of **sets**, i.e., collections of distinct objects.

Set theory: basic notation

$S = \{\dots\}$

a set and its elements

$s \in S$

s is a member of S

$s \notin S$

s is not a member of S

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Examples

The numbers 3, 5, and 9	$\{3, 5, 9\}$
The current president of the USA	$\{\text{Donald Trump}\}$
The English vowel letters	$\{a, e, i, o, u\}$
The set containing the set of even numbers and the set of odd numbers	$\big\{\{2, 4, 6, \dots\},\ \{1, 3, 5, \dots\}\big\}$

Set theory: basic notation

$S = \{\ldots\}$	$s \in S$	$s \notin S$
a set and its elements	s is a member of S	s is not a member of S

Examples

$$3 \in \{3,\ 5,\ 9\}$$

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$$7 \notin \{3,\ 5,\ 9\}$$

$$\{2,\ 4,\ 6\} \in \left\{ \{2,\ 4,\ 6\}, \mathtt{a}, \mathtt{b} \right\}$$

$$2 \notin \left\{ \{2,\ 4,\ 6\}, \mathtt{a}, \mathtt{b} \right\}$$

$$\{2,4,6\} \in ? \left\{\{\{2,\ 4,\ 6\}\}, \{\mathtt{a}\}, \mathtt{b}\right\}$$

sets can contain anything (or nothing)

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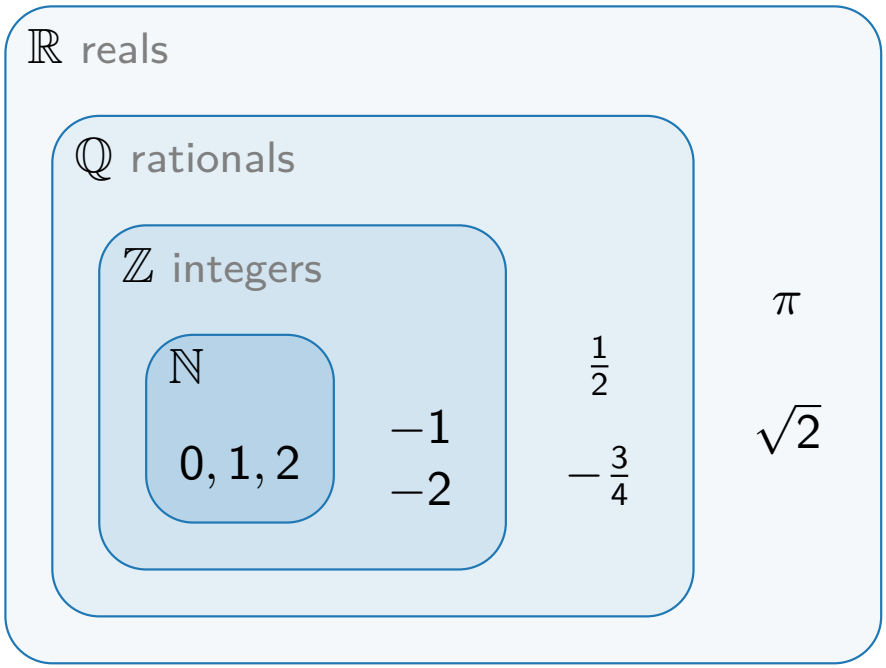
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The set containing the set of even numbers and the set of odd numbers	$\big\{\{2, 4, 6, \ldots\},\ \{1, 3, 5, \ldots\}\big\}$
The empty set, or the set that contains nothing	$\emptyset, \{\}$

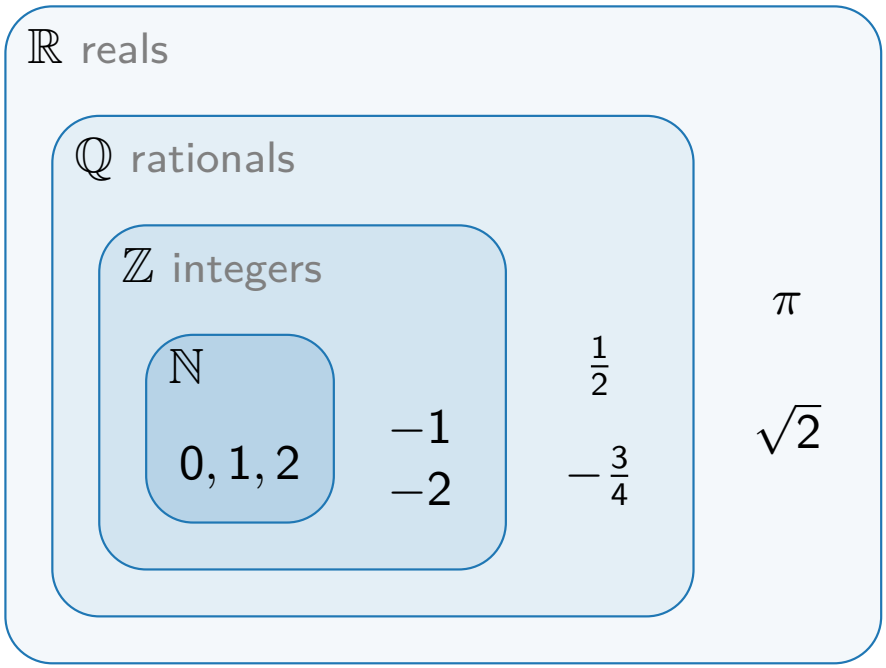
The **empty set**: $\emptyset$

Example sets studied in mathematics



In our notation

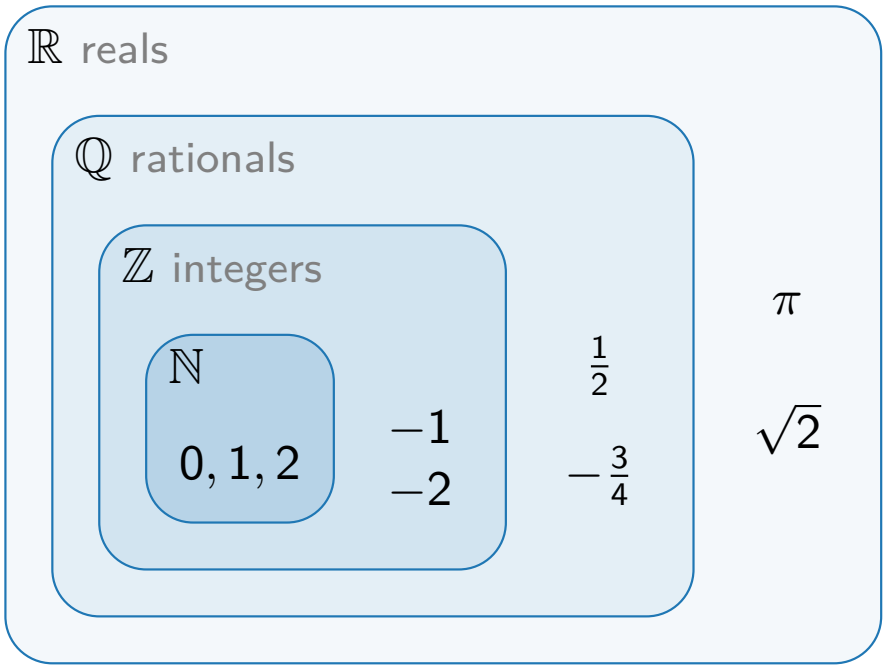
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In our notation

$\mathbb{N} = \{0, 1, 2, 3, \ldots\}$

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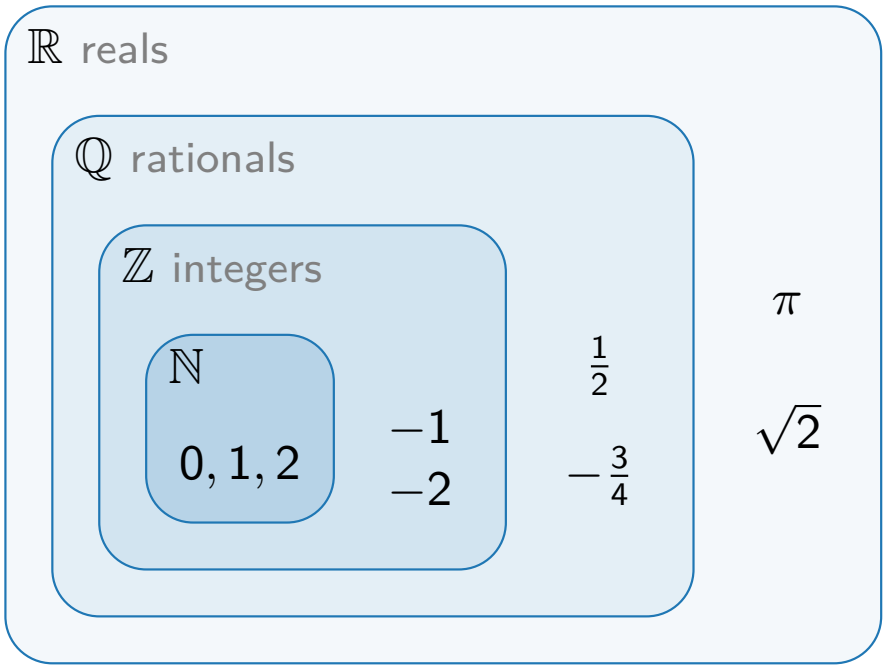


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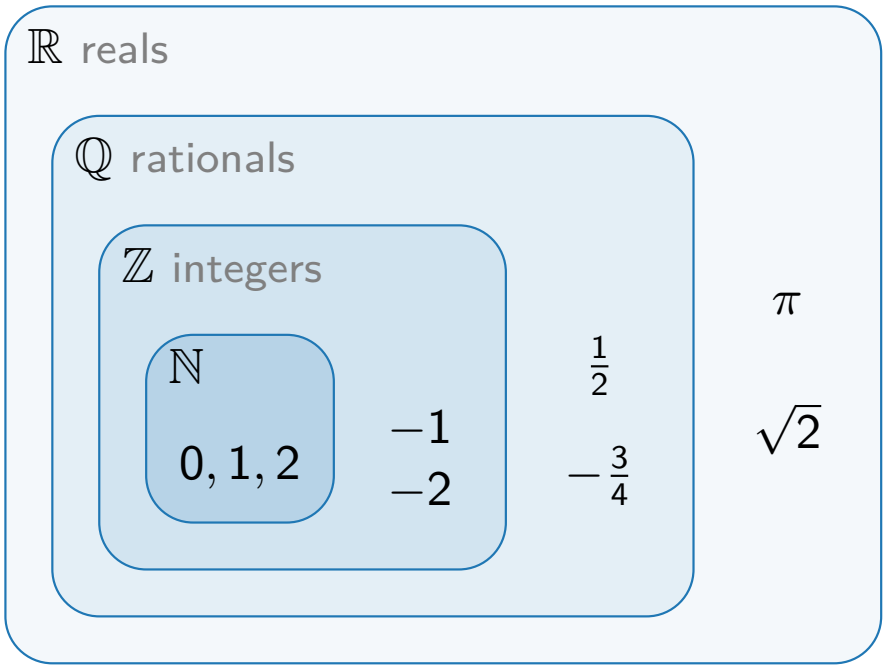
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$\mathbb{Q} = \{0, -1, \frac{1}{2}, -\frac{3}{4}, \frac{2}{3}, \ldots\}$

Example sets studied in mathematics



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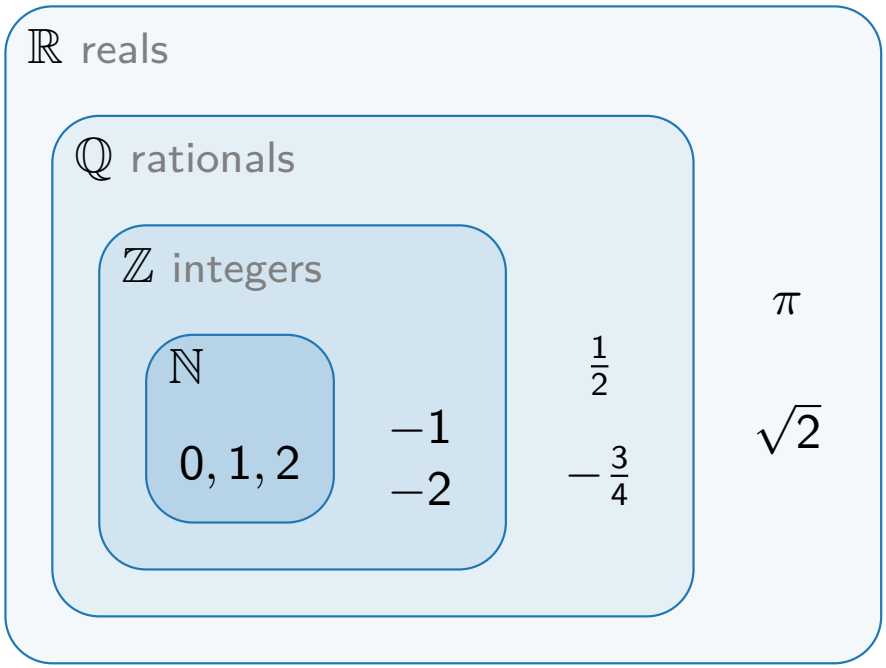
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$\mathbb{R} = \{0, -1, \frac{1}{2}, \sqrt{2}, \pi, \ldots\}$

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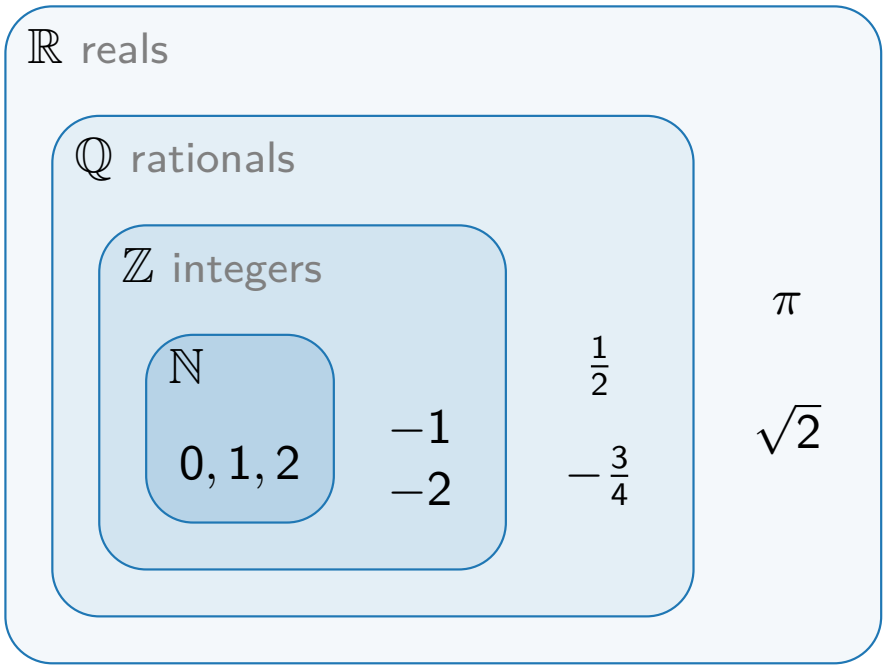
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Each set contains the ones inside it in the picture, and all four are *infinite*.

Example sets studied in mathematics



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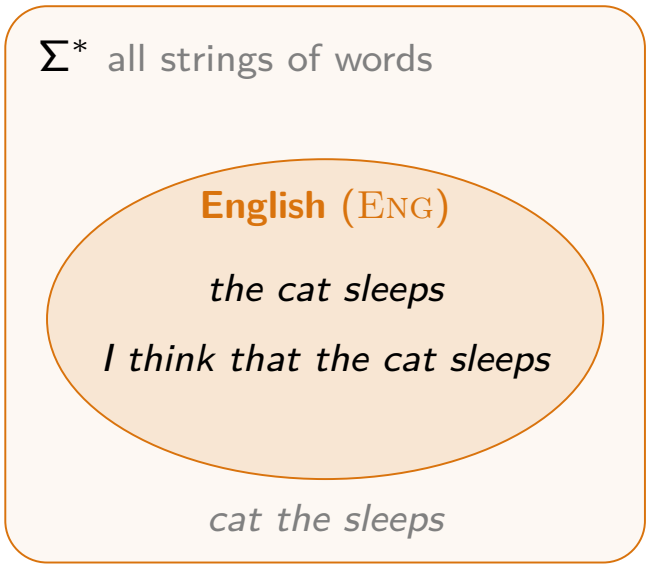
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- **Questions mathematicians ask**: what is the relationship between $\mathbb{N}$ and $\mathbb{R}$, is one set larger than the other?

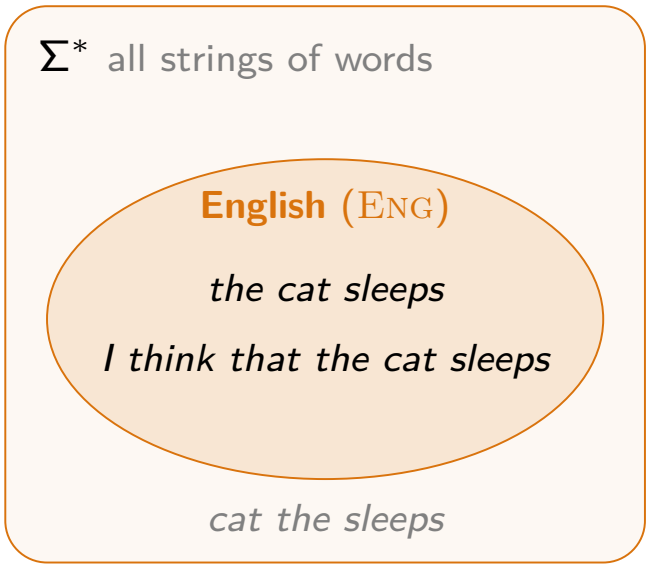
Example sets studied in computer science and linguistics



In our notation

A **language** is a set of strings.

Example sets studied in computer science and linguistics

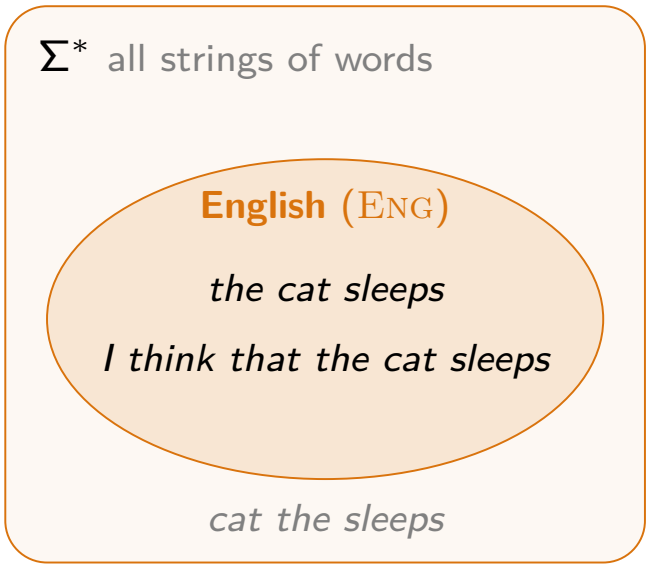


In our notation

$\Sigma = \{the, cat, sleeps, I, think, \ldots\}$

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Example sets studied in computer science and linguistics



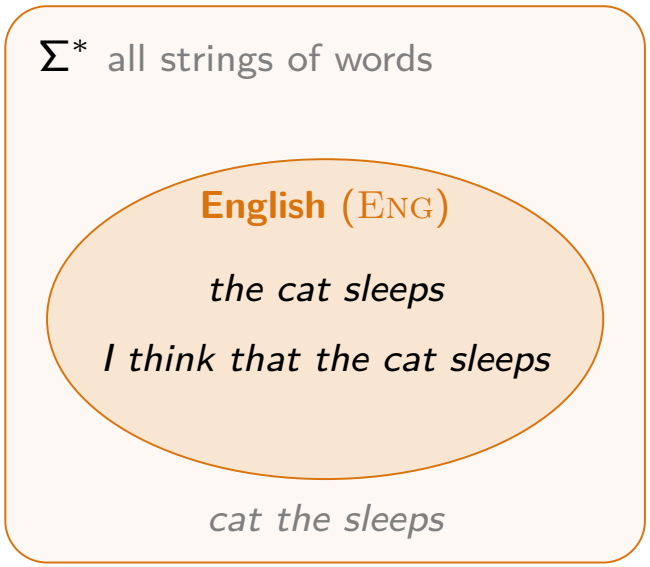
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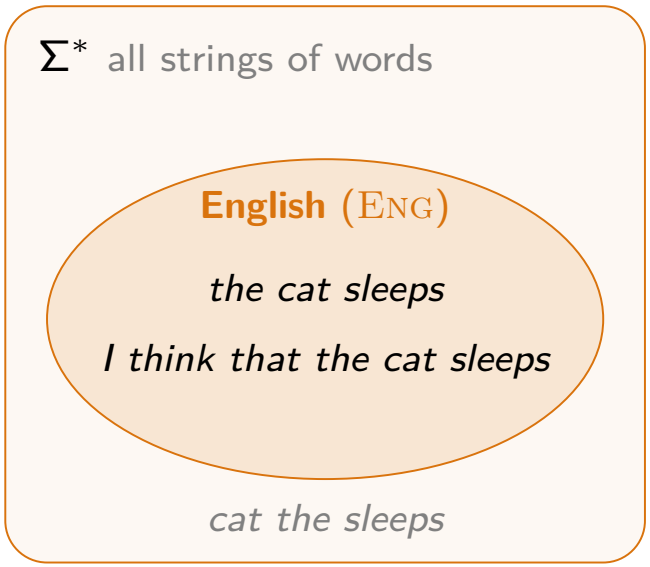
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$\text{ENG} = \{\textit{the cat sleeps}, \textit{I think that the cat sleeps}, \ldots\}$

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Example sets studied in computer science and linguistics



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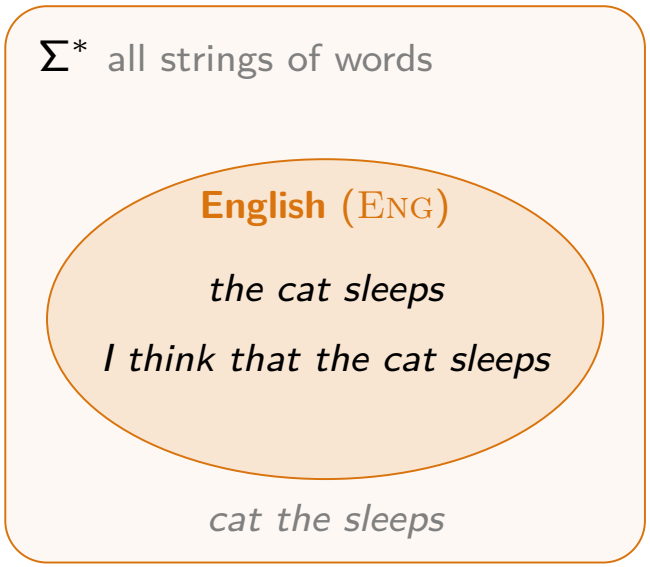
$\textit{I think that the cat sleeps}, \ldots\}$

Computer science

A programming language is also a set of strings:

$\text{PYTHON} = \{\texttt{print("hi")}, \texttt{x = 1}, \ldots\}$

Example sets studied in computer science and linguistics



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- **Questions**: How hard is it to determine if a string is in the set of Python programs? valid English expressions?

Sets in Programming (Python)

```python
# a set of numbers
S = {3, 5, 9}
print(5 in S)          # True
print(4 not in S)      # True

# the empty set
Empty = {}

# A set of english sentences
Eng = {"the cat sleeps",
       "I think that the cat sleeps"}
print("the cat sleeps" in Eng)       # True
print("cat the sleeps" in Eng)       # False
```

Fundamental: a set is determined by its elements

Two sets are **equal** exactly when they have the same elements.

$$\{a, b\} = \{b, a\}$$

order does not matter

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repetition does not matter

Sets are not lists! No duplicates

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$\{a\} \neq \{a, b\}$

b is in one but not the other

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The empty set $\emptyset = \{\,\}$

The set with no elements at all.
There is only one!

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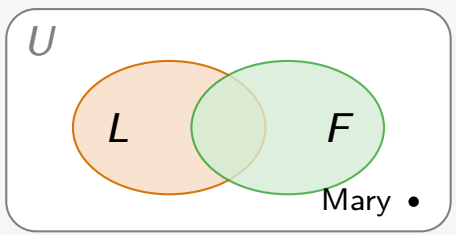
The universe U

All the entities under consideration.

The universe: what are we talking about?

The **universe** U: all the entities we are currently talking about. Every set we define lives *inside* U.

Campus

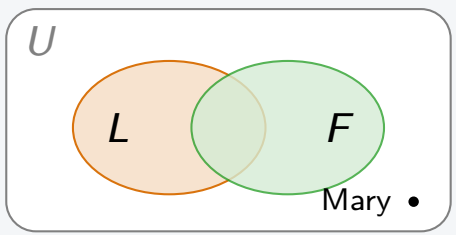


$U =$ all students

The universe: what are we talking about?

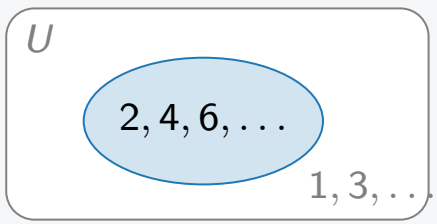
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Campus



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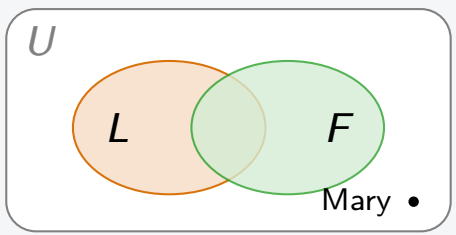


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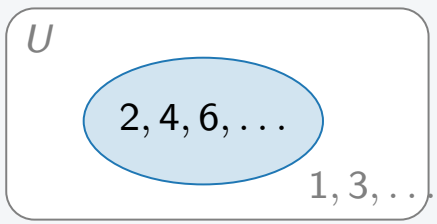
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Campus



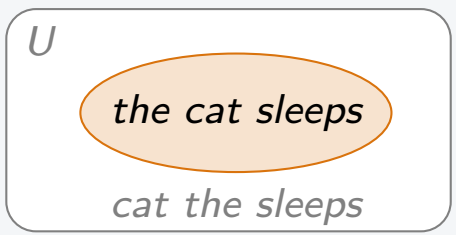
$U =$ all students

Numbers



$U = \mathbb{N}$

Language

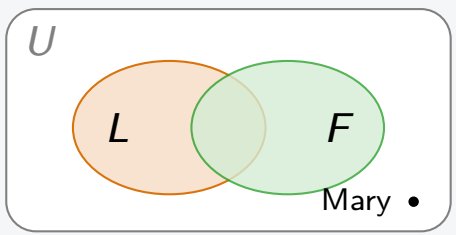


$U = \Sigma^*$

The universe: what are we talking about?

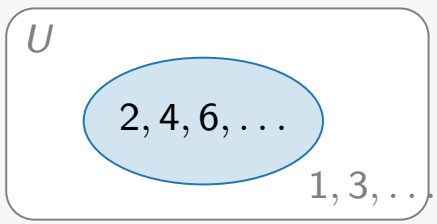
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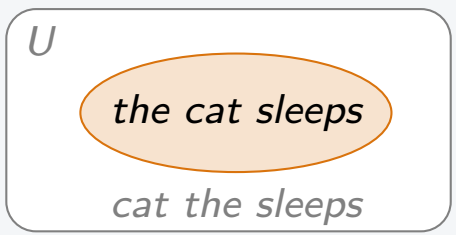
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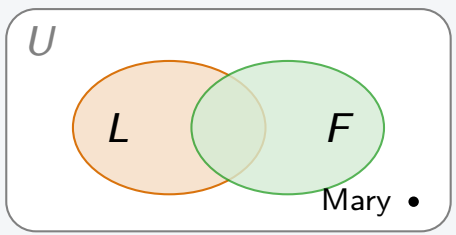
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What is *not* an even number?

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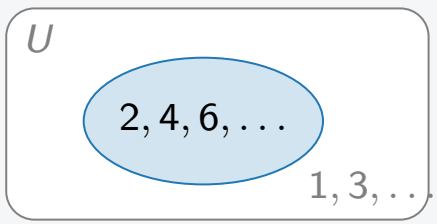
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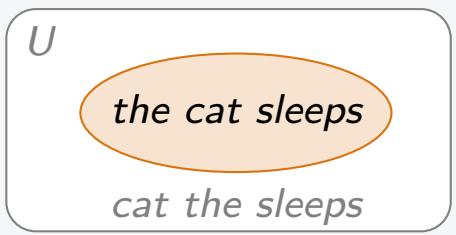
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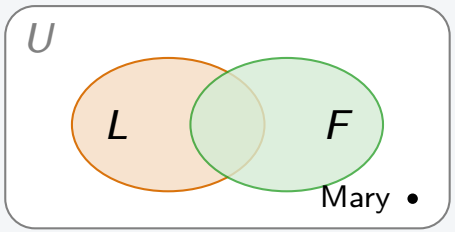
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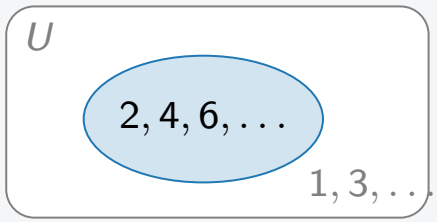
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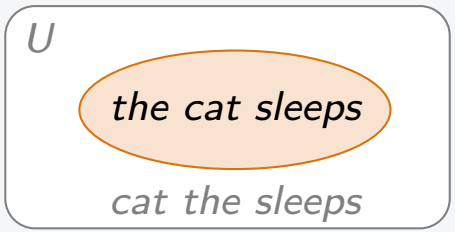
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Without a universe: also cats, chairs, Paris, ...

Interim summary

1. Sets and membership

$S = \{3, 5, 9\}$ $\quad 3 \in S$ $\quad 7 \notin S$

a set is a collection of distinct objects

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6. Sets are everywhere

$\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$ $\quad$ ENG, PYTHON

mathematics, linguistics, computer science

Quick check: true or false?

1. $\{1, 2, 3\} = \{3, 2, 1\}$
2. $\{a, a, b\}$ has three elements
3. $2 \in \{\{2, 4, 6\}, a\}$
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5.	$\{\,\} = \emptyset$	**true**	two ways to write the empty set

Things we will cover about sets

$\{\ldots\}$

How do we **define** a set?

$|X|$

How do we measure its **size**?

$X \subseteq Y$

How do we **compare** sets?

$\cup \quad \cap$

Which **operations** combine sets?

Defining sets

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3. By recursion

Give a starting point and a rule for building more.

$$0,\ n \mapsto n+1$$

1. Defining sets by listing

Examples

$X = \{2, 4, 6, 8, \ldots\}$	even integers bigger than 0
$Y = \{2, 4, 6, 8, \ldots, 20\}$	even integers no bigger than 20
$Z = \{\text{Russell}, \text{Wittgenstein}, \text{Frege}, \ldots\}$	authors to read

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The "..." only works if it is *absolutely clear* how to continue. Who else belongs in Z?

2. Defining sets by a property

Set-builder notation: read "$|$" as "*such that*"

$$\{ \underbrace{x}_{\text{elements}} \in \underbrace{\mathbb{N}}_{\text{drawn from}} \mid \underbrace{x \text{ is even}}_{\text{property}} \}$$

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The naïve part: no constraints on properties

3. Defining sets by recursion

(i) Anchor

Some things are in the set.

(ii) Step

A rule for building new elements from old ones.

(iii) Closure

Nothing else is in the set.

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A rule for building new elements from old ones.

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Nothing else is in the set.

Example: the natural numbers $\mathbb{N}$

(i) 0 is a natural number

(ii) if n is a natural number, then so is $n+1$

(iii) nothing else is a natural number

$$0 \to 1 \to 2 \to 3 \to \cdots$$

Recursion can define a language

A tiny language L

(i) number words are in L: *one*, *two*, ...

(ii) if $x, y \in L$, then so are x *plus* y and x *minus* y

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Which strings are in L?

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twenty minus three

twenty minus three plus four

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Finite rules, infinitely many sentences. We will define the formulas of logic the same way.

The size of a set

$\{\ldots\}$

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The **cardinality** $|X|$ of a set X is the number of elements in X.

$|\{a\}| = 1$

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- Might seem trivial, but very interesting results emerge when you start talking about infinite sets!

Comparing sets

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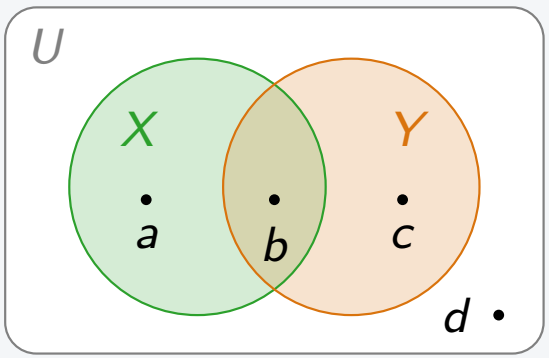
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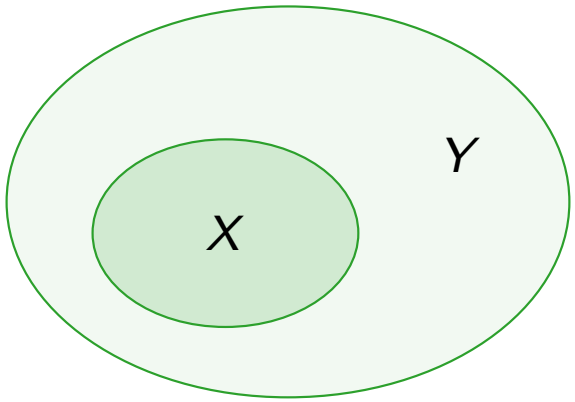
Comparing sets

A tool for comparing sets: Venn diagrams



rectangle	the universe U
circle	a set, e.g., X
dot	an element
inside	a member: $a \in X$
overlap	in both X and Y: b
outside	in neither: d

Subsets



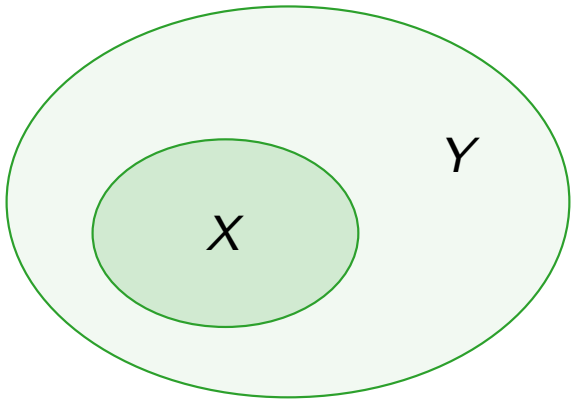
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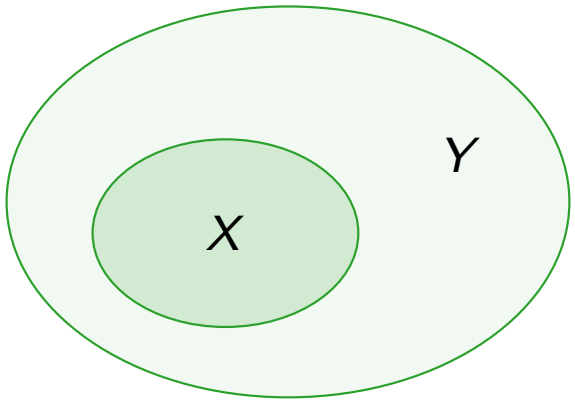
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Examples

$\{a, b\} \subseteq \{a, b, c\}$ $\quad\quad$ $\mathbb{N} \subseteq \mathbb{Z}$

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How do we check whether $X \subseteq Y$?

To show $X \subseteq Y$

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$\{a, b\} \subseteq_? \{a, b\}$	a ✓	b ✓		**yes**	true for every set

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$\{a, b\} \subset_? \{a, b, c\}$

$\{a, b\} \subset_? \{a, b\}$

$\mathbb{N} \subset_? \mathbb{Z}$

Proper subsets

Definition

$X \subset Y$ (X is a **proper subset** of Y)

if $X \subseteq Y$ **and** Y has at least one element that is not in X.

To show $X \subset Y$

1. show $X \subseteq Y$
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$\subseteq$ is like $\leq$ $\subset$ is like $<$

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$\{a, b\} \subset_? \{a, b, c\}$ **yes** $\{a, b\} \subseteq \{a, b, c\}$, and c is extra

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$\{a, b\} \subset_? \{a, b, c\}$	**yes**	$\{a, b\} \subseteq \{a, b, c\}$, and c is extra
$\{a, b\} \subset_? \{a, b\}$	**no**	nothing extra (but $\{a, b\} \subseteq \{a, b\}$)
$\mathbb{N} \subset_? \mathbb{Z}$	**yes**	e.g., -1 is extra

An important fact about the empty set

The empty set is a subset of **every** set:

$$\emptyset \subseteq X \quad \text{for every set } X$$

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$$\emptyset \subseteq \mathbb{N}$$

$$\emptyset \subseteq \emptyset$$

Why? (intuition)

To show $\emptyset \not\subseteq X$, we would need a counterexample: an element of $\emptyset$ that is not in X. But $\emptyset$ has no elements, so there is *nothing that could go wrong*.

We will prove this formally later in the course.

Operations on sets

$\{\ldots\}$

How do we **define** a set?

$|X|$

How do we measure its **size**?

$X \subseteq Y$

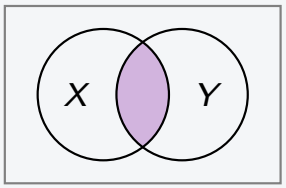
How do we **compare** sets?

$\cup \quad \cap$

Which **operations** combine sets?

Combining sets

Intersection

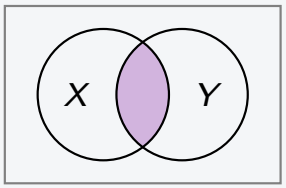


$X \cap Y$

elements in X
and in Y

Combining sets

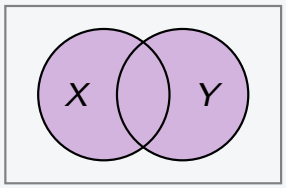
Intersection



$X \cap Y$

elements in X **and** in Y

Union

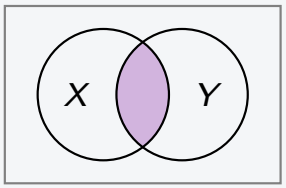


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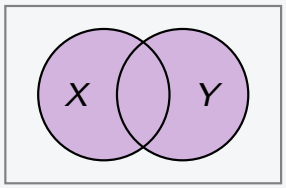
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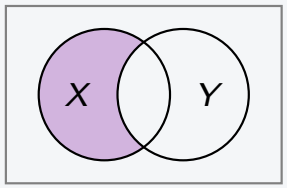
Union



$X \cup Y$

elements in X **or** in Y

Difference

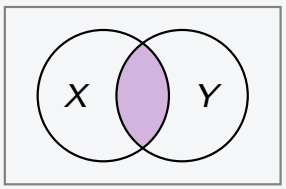


$X \setminus Y$

elements in X and **not** in Y

Combining sets

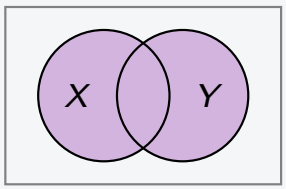
Intersection



$X \cap Y$

elements in X **and** in Y

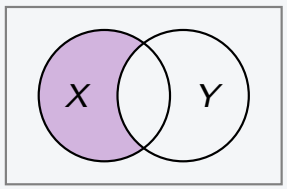
Union



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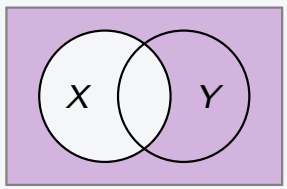
Difference



$X \setminus Y$

elements in X and **not** in Y

Complement

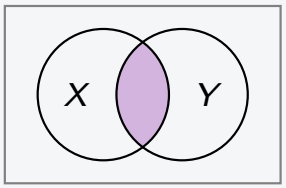


$\overline{X}$

elements in U and **not** in X

Combining sets

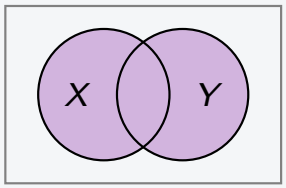
Intersection



$X \cap Y$

elements in X **and** in Y

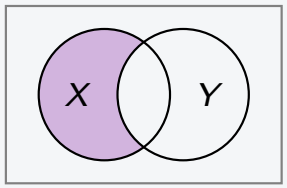
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elements in X **or** in Y

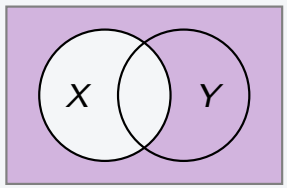
Difference



$X \setminus Y$

elements in X and **not** in Y

Complement



$\overline{X}$

elements in U and **not** in X

In set-builder notation

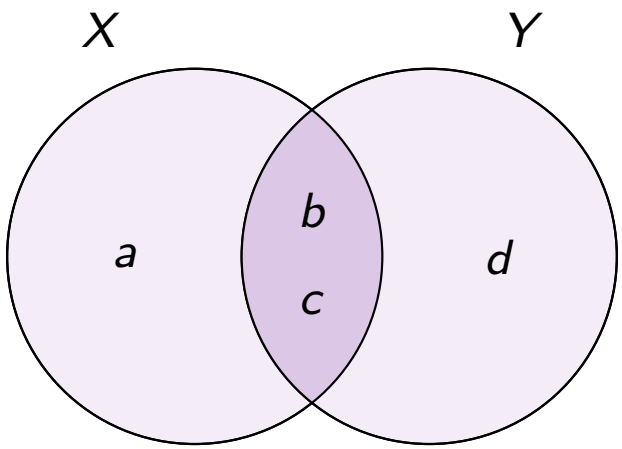
$X \cap Y = \{z \mid z \in X \text{ and } z \in Y\}$

$X \cup Y = \{z \mid z \in X \text{ or } z \in Y\}$

$X \setminus Y = \{z \mid z \in X \text{ and } z \notin Y\}$

$\overline{X} = \{z \in U \mid z \notin X\}$

Combining sets: examples



$X = \{a, b, c\}$ $Y = \{b, c, d\}$

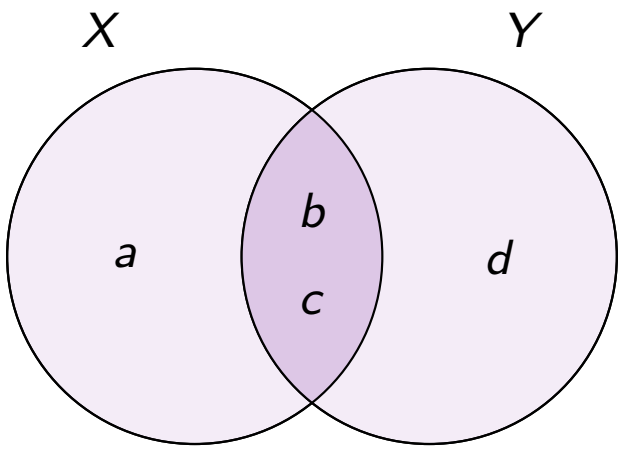
$X \cap Y =$

$X \cup Y =$

$X \setminus Y =$

$Y \setminus X =$

Combining sets: examples



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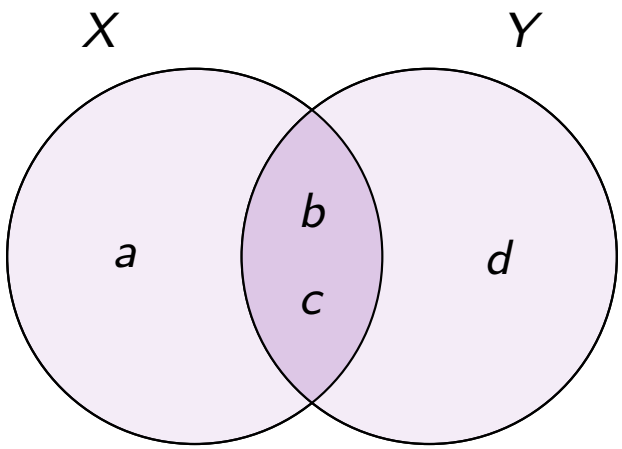
$X \cap Y = \{b, c\}$

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$X \setminus Y =$

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Combining sets: examples



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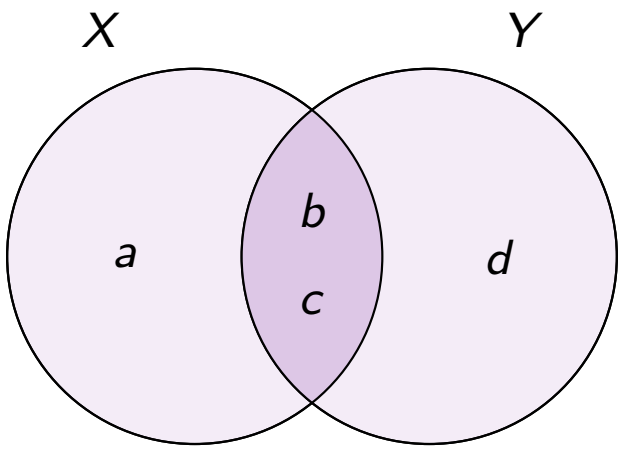
$X \cap Y = \{b, c\}$

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$X \setminus Y =$

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Combining sets: examples



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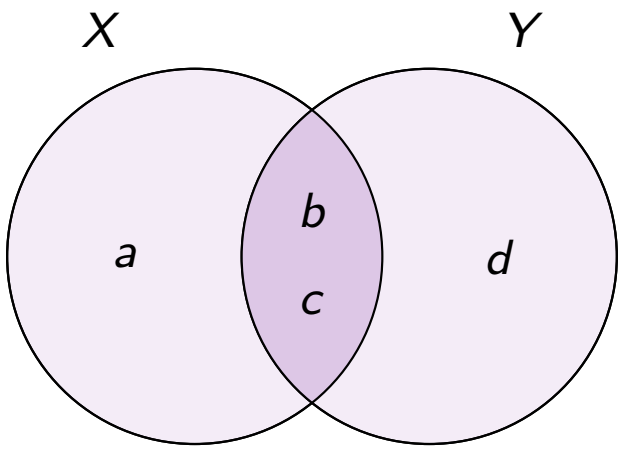
$X \cap Y = \{b, c\}$

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Combining sets: examples



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$X \cap Y = \{b, c\}$

$X \cup Y = \{a, b, c, d\}$

$X \setminus Y = \{a\}$

$Y \setminus X = \{d\}$

Combining sets: examples

Set operations in Python

```
X = {"a", "b", "c"}
Y = {"b", "c", "d"}

# union, intersection, difference
print(X | Y)    # {'a','b','c','d'}
print(X & Y)    # {'b','c'}
print(X - Y)    # {'a'}
print(Y - X)    # {'d'}
# (printed order may vary)

print(len(X))       # cardinality: 3
print({"a"} <= X)   # subset: True
```

Some laws that follow from the definitions

Commutative

$$X \cup Y = Y \cup X$$

$$X \cap Y = Y \cap X$$

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$$X \cup Y = Y \cup X$$
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Associative

$$(X \cup Y) \cup Z = X \cup (Y \cup Z)$$
$$(X \cap Y) \cap Z = X \cap (Y \cap Z)$$

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$$(X \cup Y) \cup Z = X \cup (Y \cup Z)$$
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Distributive

$$X \cup (Y \cap Z) = (X \cup Y) \cap (X \cup Z)$$
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Some laws that follow from the definitions

Commutative

$$X \cup Y = Y \cup X$$
$$X \cap Y = Y \cap X$$

Identity

$$X \cup \emptyset = X \qquad X \cup U = U$$
$$X \cap U = X \qquad X \cap \emptyset = \emptyset$$

Associative

$$(X \cup Y) \cup Z = X \cup (Y \cup Z)$$
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Complement

$$X \cup \overline{X} = U \qquad X \cap \overline{X} = \emptyset$$
$$\overline{\overline{X}} = X \qquad X \setminus Y = X \cap \overline{Y}$$

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Complement

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How do we *know* these are true? We need **proofs** (discussed later).

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These should become second nature

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An important concept: the power set

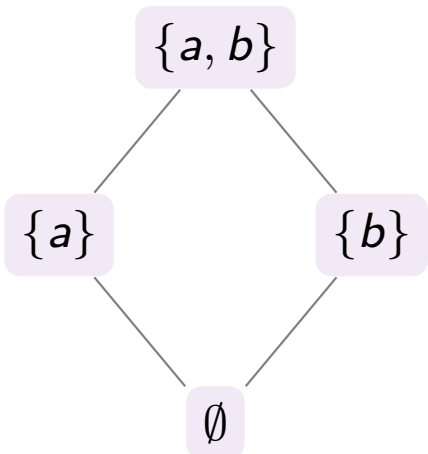
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lines go up from a set to a bigger set containing it

Example

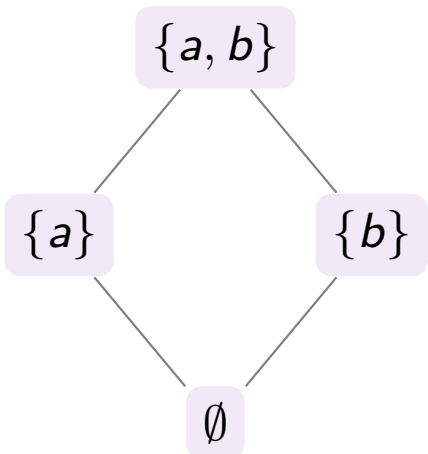
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Example

$X = \{a, b\}$

$\mathcal{P}(X) = \{\emptyset, \{a\}, \{b\}, \{a, b\}\}$

Cardinality (finite sets)

$|\mathcal{P}(X)| = 2^{|X|} = 2^2 = 4$

for each element: in or out?

Try this yourself, be mindful of the empty set!

Conclusion: what you should know

Defining sets

- ✓ notation: $\{\ldots\}$, $\in$, $\notin$, $\emptyset$, U
- ✓ $\{a, b\} = \{b, a\} = \{a, a, b\}$
- ✓ listing, set-builder, recursion

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- ✓ check $X \subseteq Y$ and $X \subset Y$
- ✓ find a counterexample when not
- ✓ $\emptyset \subseteq X$ for every X
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Next time: relations and functions, built out of sets.